上海交通大學
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Carbon Accounting

Lecture 5

Carbon Accounting: National Inventory

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China's Carbon-Intensity Target: Achieved Three Years Early?

How would you verify this claim?



Reuters World Business Markets Sustainability Legal More

China says to cut carbon intensity 40 to 45 percent by 2020

By **Reuters**

November 26, 2009 4:17 PM GMT+8 · Updated November 26, 2009

BEIJING (Reuters) - China will reduce its carbon intensity -- the amount of carbon dioxide emitted for each unit of GDP -- 40 to 45 percent by 2020, compared with 2005 levels, the official Xinhua news agency reported on Thursday.

"This is a voluntary action taken by the Chinese government based on its own national conditions and is a major contribution to the global effort in tackling climate change," Xinhua said, quoting the State Council or cabinet.

Reporting by Emma Graham-Harrison; Editing by David Fogarty

Reuters World Business Markets Sustainability Legal More

China meets 2020 carbon target ahead of schedule: Xinhua

By **Reuters**

March 27, 2018 6:36 PM GMT+8 · Updated March 27, 2018



Baseline: 2005

Target year: 2020

Target: carbon intensity −40% to −45%

How Would You Verify This Claim?

China reported that it had achieved its 2020 carbon-intensity target three years early. What data would you need, and what calculation would you make, to verify this claim?

Answer in 1–2 sentences.

You do not need to provide actual numbers.

作答

How Can We Verify the Claim?

What data do we need?

- China's CO_2 emissions in 2005
- China's GDP in 2005
- China's CO_2 emissions in 2017
- China's GDP in 2017

What do we calculate?

- Calculate the carbon intensity for both years.
- Calculate the percentage reduction from 2005 to 2017.
- Compare the result with the 40%–45% target.

Carbon intensity:

$$CI_{2005} = \frac{CO_{2,2025}}{GDP_{2005}}$$

$$CI_{2017} = \frac{CO_{2,2017}}{GDP_{2017}}$$

Percentage reduction:

$$Reduction = \frac{CI_{2005} - CI_{2017}}{CI_{2005}} \times 100\%$$

GDP comes from the national accounts.
But where does the national CO_2 emissions figure come from?

Learning Outcomes

1. **Explain** why national GHG inventories emerged and identify their major reporting and policy purposes.
2. **Describe** the coverage and structure of a national GHG inventory.
3. **Distinguish** production- and consumption-based accounting..
4. **Analyze** the comparability and reliability of national emissions estimates.

目 录

Contents

01- Why National GHG Inventories?

National inventories in international climate governance

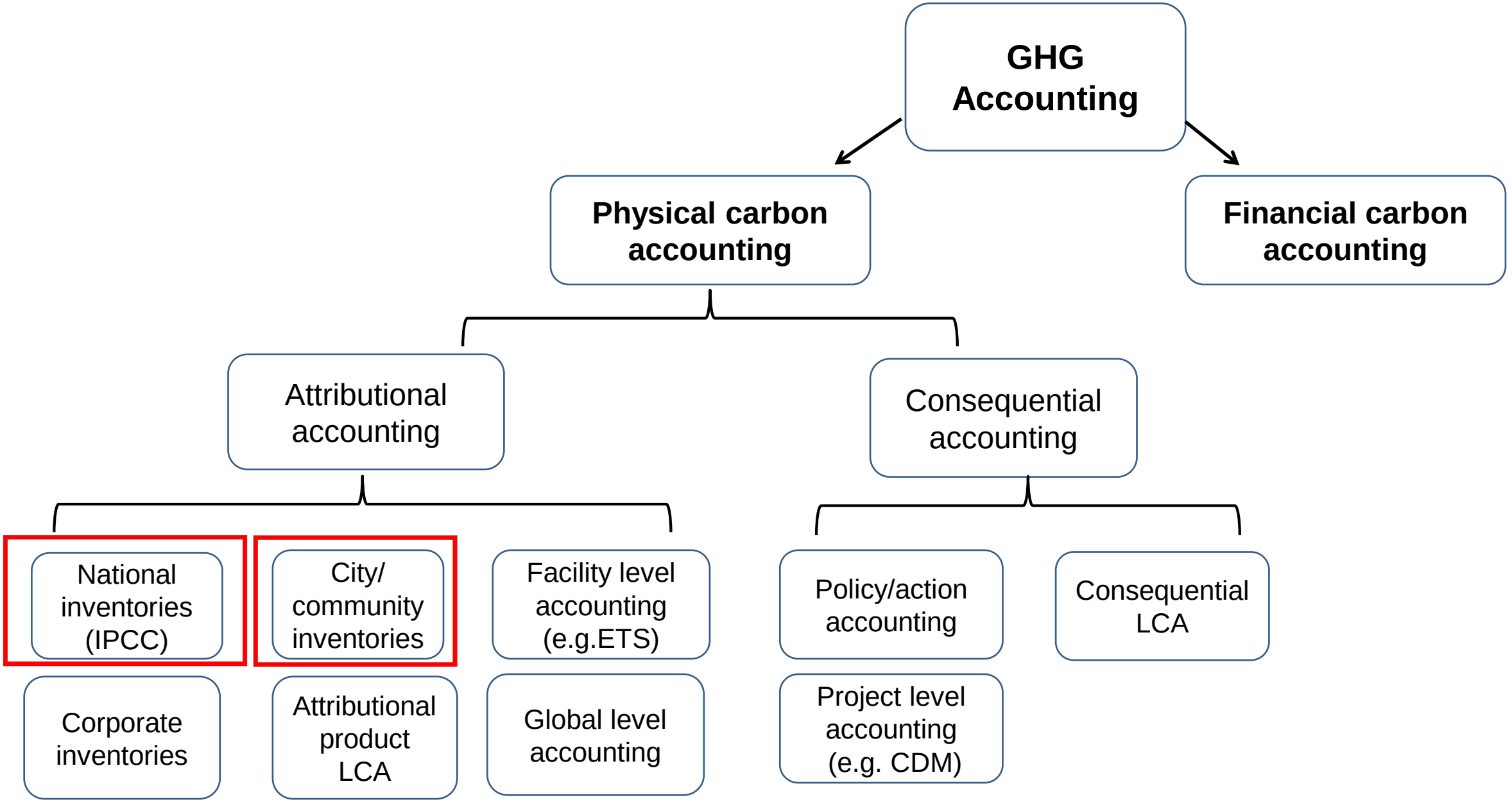
02- What is Counted and How?

Guidelines, Boundaries, emission categories, data and calculation

03- How comparable and reliable are the results?

Production vs consumption accounting and the gigatonne gap

04 – Wrap up and Next time


GHG Accounting
Physical carbon accounting
Financial carbon accounting
Attributional accounting
Consequential accounting
National inventories (IPCC)
City/ community inventories
Facility level accounting (e.g.ETS)
Policy/action accounting
Consequential LCA
Corporate inventories
Attributional product LCA
Global level accounting
Project level accounting (e.g. CDM)

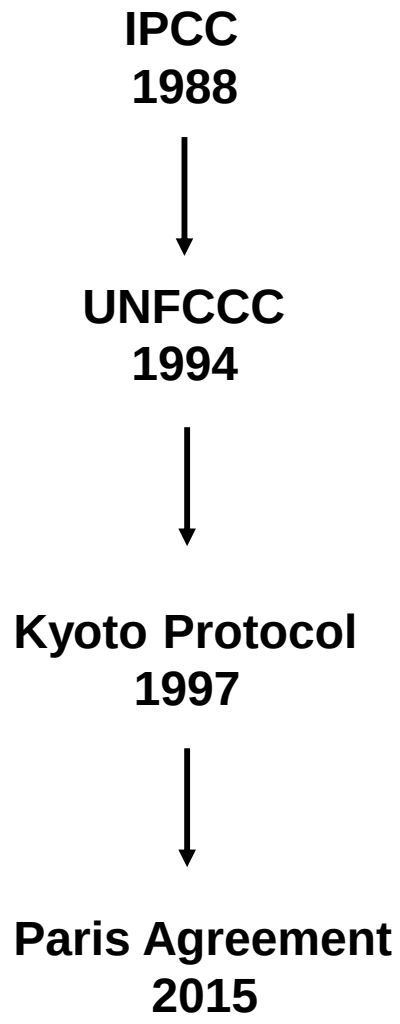
01

Why National GHG Inventories?

Why National GHG Inventories Came First

- National GHG inventories emerged from the need to support international climate governance.

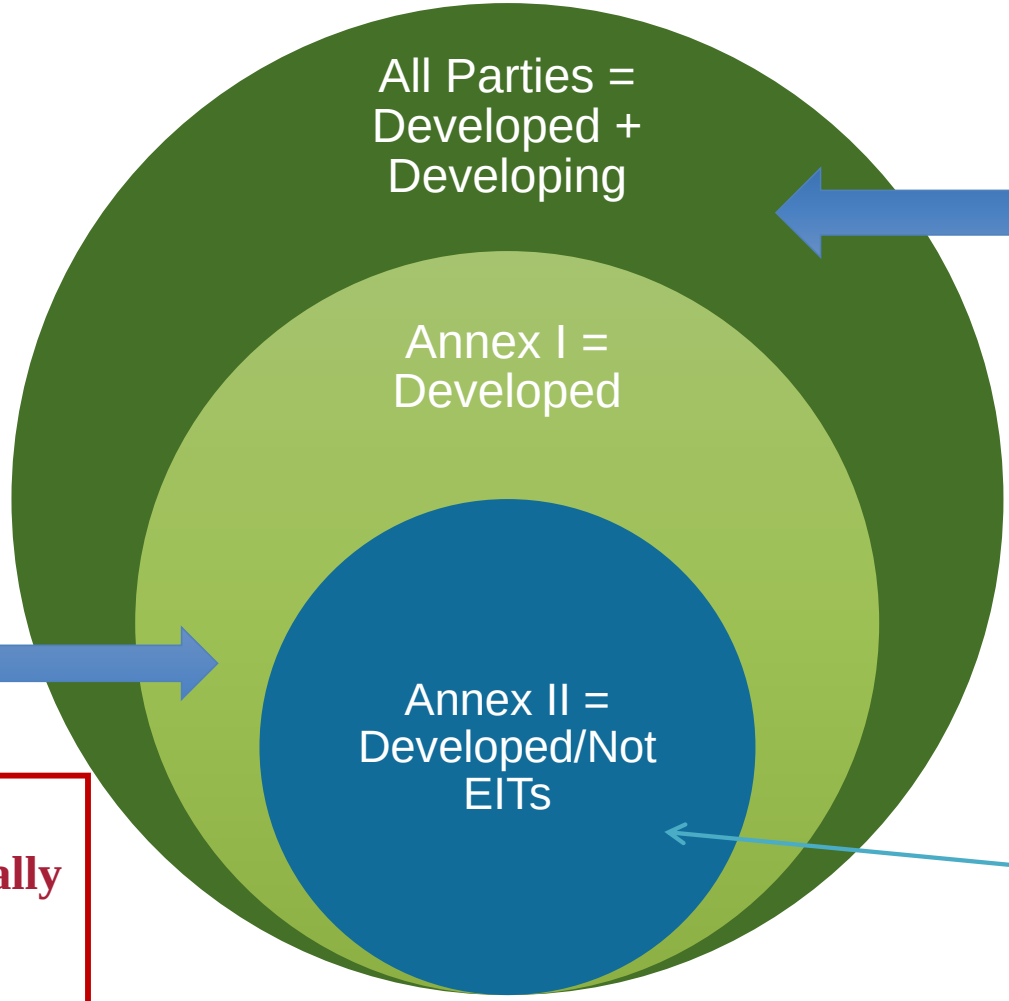
1. **Climate change is a global collective-action problem**
   Emissions occur across countries, but impacts are global.
2. **Countries are the primary actors in international climate agreements.**
   Commitments, reporting, and review are organized at the national level.
3. **Climate governance requires comparable emissions information**
   Countries need a consistent way to estimate and report anthropogenic GHG emissions and removals.
4. **National inventories provide the data basis**
   They support reporting, target tracking, transparency, and policy assessment.



Some context

The UNFCCC divides countries into three main groups:

• Annex I Parties • Annex II Parties • Non-Annex I Parties



All Parties:

- **National Emissions Inventory**
- Develop programs
- Cooperate in tech transfer
- Manage sinks and reservoirs

Subject to "common but Differentiated responsibilities"

Annex I:

- **Emissions inventory annually**
- Develop national programs
- And take actions to mitigate

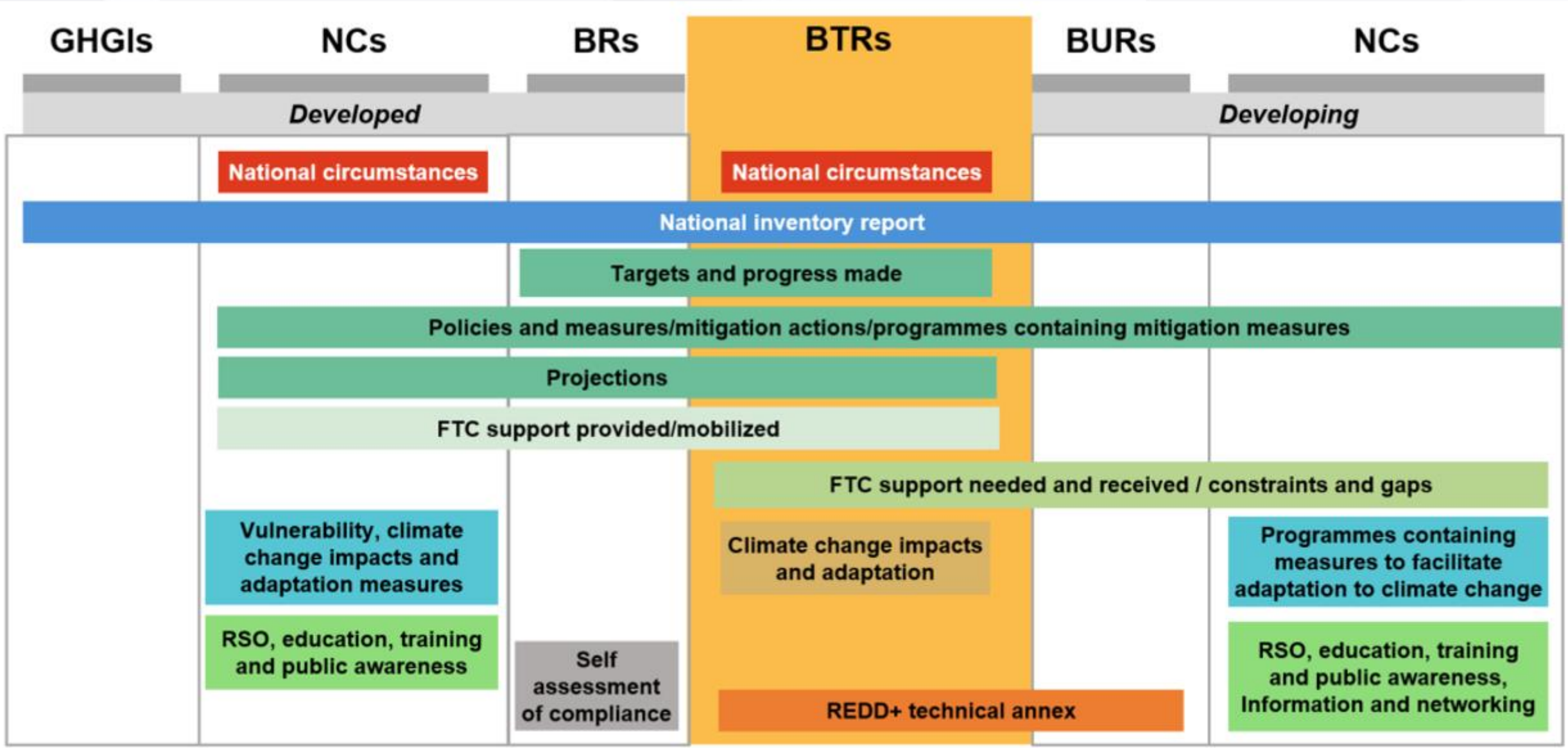
Annex II:

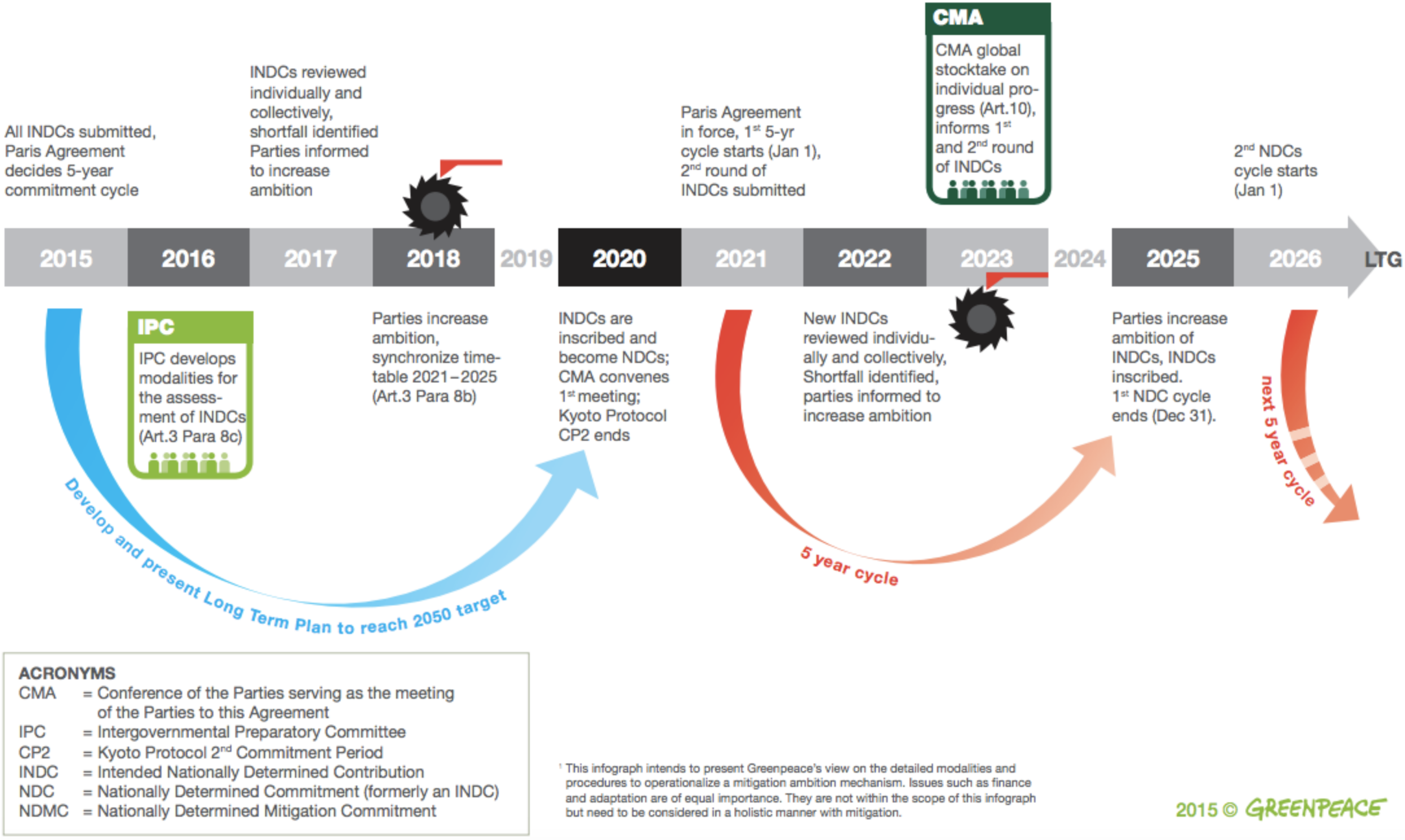
- Transfer technology
- Fund adaptation

NC, BUR and BTR – Non-Annex countries

- National Communications (NC) and Biennial Update Report (BUR) are required for Non-Annex I countries using comparable methodologies and consistent with their capabilities for reporting. (GHG inventory as part of the NC and BUR)
  - List of all submissions of NC https://unfccc.int/non-annex-I-NCs
  - List of all submissions of BUR https://unfccc.int/BURs

- Transitioning to Biennial Transparency Reports (BTR) for Paris Agreement Parties
  - The first BTR is due on 31 December 2024.
  - List of all submissions of BTR https://unfccc.int/first-biennial-transparency-reports
  - China's BTR


GHGIs
NCs
BRs
BTRs
BURs
NCs
Developed
Developing
National circumstances
National circumstances
National inventory report
Targets and progress made
Policies and measures/mitigation actions/programmes containing mitigation measures
Projections
FTC support provided/mobilized
FTC support needed and received / constraints and gaps
Vulnerability, climate change impacts and adaptation measures
Climate change impacts and adaptation
Programmes containing measures to facilitate adaptation to climate change
RSO, education, training and public awareness
Self assessment of compliance
REDD+ technical annex
RSO, education, training and public awareness, Information and networking



Source: [carbon brief]

Discussion

What does a national GHG inventory enable?

Report **Track** **Plan** **Improve** **Communicate** **…**

02

What Is Counted – and How?

Guidelines, Boundaries, emission categories, data and calculation

What a national inventory looks like under UNFCCC…

- http://di.unfccc.int/time_series (GHG data interface)

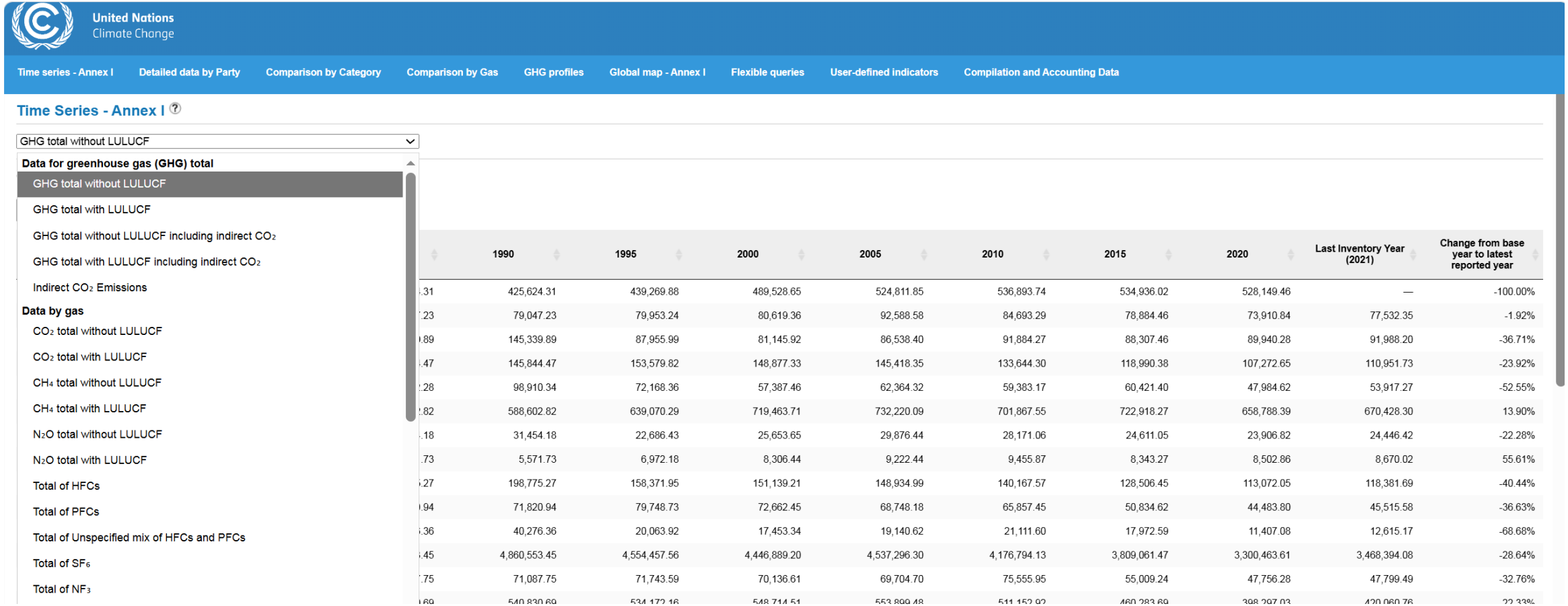


Table ES 3 **Aggregated emission trends per source category, including all estimated GHG emissions from the Crown Dependencies and selected relevant Overseas Territories (Mt CO_2 equivalent)**

Source Category	1990	1995	2000	2005	2010	2014	2015	2016	2017	2018
1. Energy	614.9	572.8	565.9	562.4	507.7	432.2	416.0	393.8	382.4	375.3
2. Industrial Processes and Product Use	66.8	61.2	40.9	40.0	35.8	34.4	33.9	31.0	30.6	28.5
3. Agriculture	49.2	48.0	46.0	43.6	40.9	41.8	41.2	41.2	41.6	41.2
4. LULUCF	0.1	-2.1	-3.9	-7.0	-8.9	-9.4	-9.6	-9.6	-9.8	-10.0
5. Waste	67.0	69.7	63.4	49.4	30.0	21.5	21.0	20.4	20.7	21.0
Total (net emissions)	797.9	749.6	712.4	688.5	605.6	520.6	502.5	476.8	465.5	456.0

'Attributes'

Footnotes: Geographical coverage of this table includes the Crown Dependencies and the Overseas Territories which are included in the scope of the UK's ratification of the UNFCCC

Source: United Kingdom. 2020 National Inventory Report (NIR)
https://unfccc.int/documents/225987

What information can you find?

Table 1-7 China's Total GHG Emissions in 2020 ($MtCO_2eq$)

	CO_2	CH_4	N_2O	HFCs	PFCs	SF_6	Total
Energy	9,661	782	117				10,560
IPPU	1,532	0	130	273	22	93	2,049
Agriculture		664	251				915
LULUCF	-1,343	39	0				-1,303
Waste	9	207	27				242
Total (without LULUCF)	11,202	1,653	524	273	22	93	13,766
Total (with LULUCF)	9,859	1,692	524	273	22	93	12,463

Notes: 1). Shaded cells do not require entries;

2). 0 indicates that the value is less than 0.5.

3). Due to rounding, the aggregation of various items may be slightly different from the total.

Source : “The People’s Republic of China Fourth Biennial Update Report on Climate Change”(2024)

Table 2-1 GHG Emissions and Removals of China in 2010 (Mt CO_2 eq)

	CO_2	CH_4	N_2O	HFCs	PFCs	SF_6	Total
Total (with LULUCF)	7,678	1,163	547	132	10	21	9,551
1. Energy	7,624	564	96				8,283
2. Industrial processes	1,075	0	62	132	10	21	1,301
3. Agriculture		471	358				828
4. LULUCF	-1,030	37	0				-993
5. Waste	8	92	31				132
Total (without LULUCF)	8,707	1,127	547	132	10	21	10,544

Note: 1. Shaded cells do not require entries;

From:
China Third National Communication on Climate Change (2019)

Table 2-7 GHG Emissions and Removals of China in 2014（Mt CO_2 eq）

	CO_2	CH_4	N_2O	HFCs	PFCs	SF_6	Total
1. Energy	8,925	520	114				9,559
2. Industrial processes	1,330	0	96	214	16	61	1,718
3. Agriculture		467	363				830
4. Waste	20	138	37				195
5. LULUCF	-1,151	36	0				-1,115
Total (without LULUCF)	10,275	1,125	610	214	16	61	12,301
Total (with LULUCF)	9,124	1,161	610	214	16	61	11,186

Note: 1. Shaded cells do not require entries;

2. 0 indicates that the calculation result is less than 0.5 Mt CO_2 eq;

3. Due to rounding, the aggregation of various items may be slightly different from the total.

From:
China Second Biennial Update Report on Climate Change (2019)

National Inventories: methodology

- IPCC guidance:

http://www.ipcc-nggip.iges.or.jp/public/2006gl/

- IPCC LULUCF guidance:

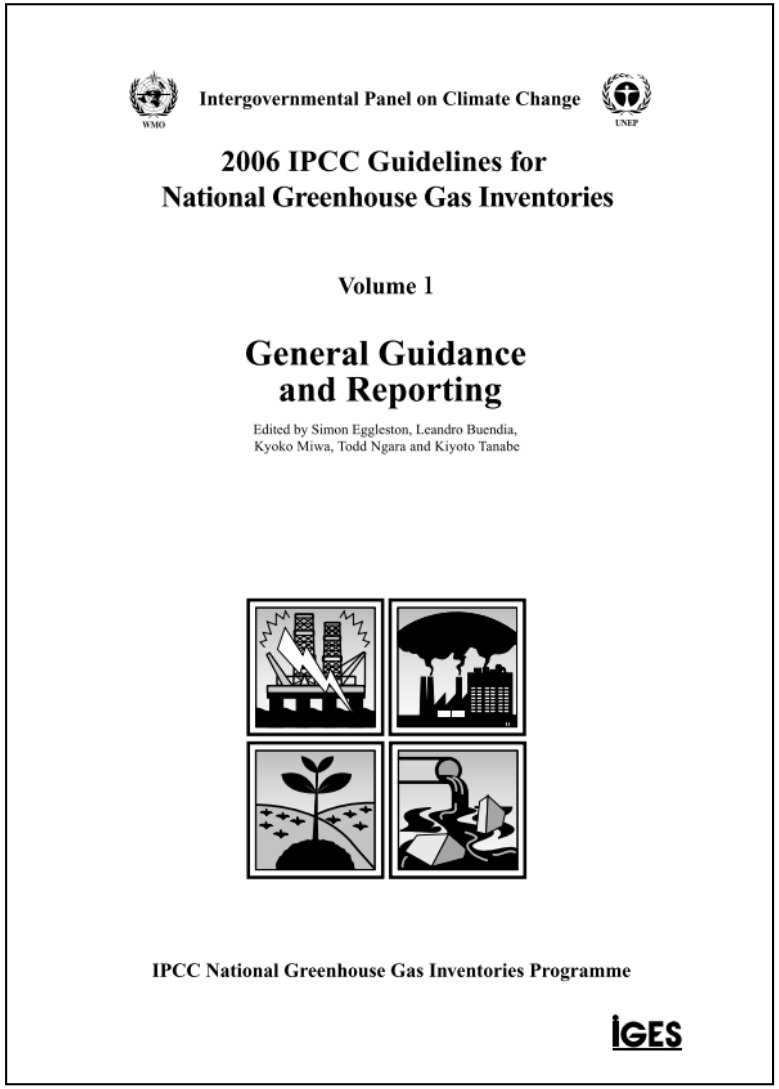
http://www.ipcc-nggip.iges.or.jp/public/gpglulucf/gpglulucf.html

- IPCC 2013 Update:

http://www.ipcc-nggip.iges.or.jp/public/kpsg/ (for Kyoto Protocol)

- IPCC guidance 2019 Refinement

https://www.ipcc.ch/report/2019-refinement-to-the-2006-ipcc-guidelines-for-national-greenhouse-gas-inventories/



IPCC Inventory Guidelines and UNFCCC

New Supplementary Guidance in 2013



Currently, Non Annex I Parties use these under the UNFCCC.

Annex I Parties must use from 2015

Non-Annex I Parties are encouraged to use GPGs.

GPG2000 (non-LULUCF)

GPG2003 (LULUCF)

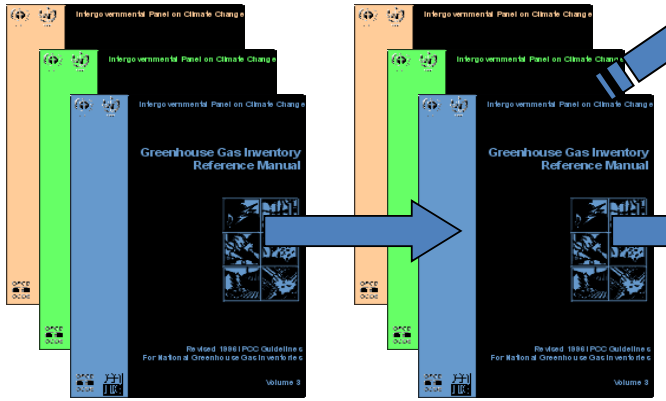


2006 IPCC Guidelines

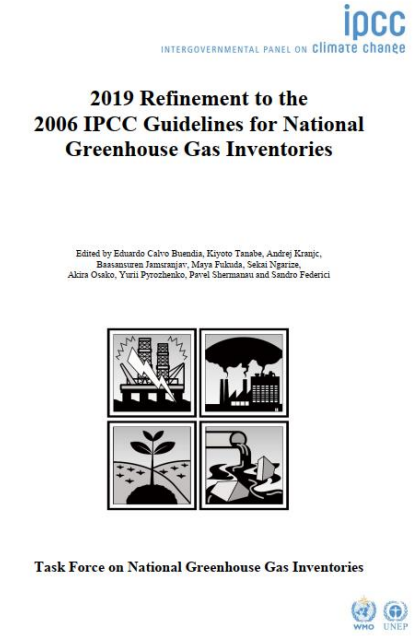


1995 IPCC Guidelines

Revised 1996 IPCC Guidelines



Actually, *2006 IPCC Guidelines* are being used by more and more Non-Annex I Parties.



Revision/Update by the IPCC

2019 Refinement

IPCC Guidelines

- **2006 IPCC Guidelines**: core methodological framework
- **2019 Refinement**: updates, supplements, and elaborations
- Some key concepts:
  - Anthropogenic emissions and removals
  - National territory
  - Inventory year and time series
  - Inventory reporting
  - Greenhouse gases
  - Other gases
  - **Sectors and Categories**
  - **Tiered methods**

Sectoral Structure in IPCC Guidelines

- Major inventory sectors:
  1. Energy
  2. Industrial Processes and Product Use (IPPU)
  3. Agriculture, Forestry and Other Land Use (AFOLU)
  4. Waste
  5. Other

Energy

A. Fuel Combustion Activity
B. Fugitive Emissions from Fuel
C. CO_2 Transport and Storage

AFOLU

A. Livestock
B. Land
C. Aggregate Sources and Non-CO_2 Emission Source on Land

Waste

A. Solid Waste Disposal
B. Biological Treatment of Solid Waste
C. Incineration and Open Burning of Waste
D. Wastewater Treatment and Discharge

IPPU

A. Mineral Industry
B. Chemical Industry
C. Metal Industry
D. Non-Energy Products from Fuel and Solvent Use
E. Electronics Industry
F. Product Use as Substitutes for ODS
G. Other Product Manufacture and Use
H. Other

- 0_Overview
- 1_Volume1
- 2_Volume2
- 3_Volume3
- 4_Volume4
- 5_Volume5

- V1_0_Cover.pdf
- V1_1_Ch1_Introduction.pdf
- V1_2_Ch2_DataCollection.pdf
- V1_3_Ch3_Uncertainties.pdf
- V1_4_Ch4_MethodChoice.pdf
- V1_5_Ch5_Timeseries.pdf
- V1_6_Ch6_QA_QC.pdf
- V1_7_Ch7_Precursors_Indirect.pdf
- V1_8_Ch8_Reporting_Guidance.pdf
- V1_8x_Ch8_An1_Units_Index.pdf
- V1_8x_Ch8_An2_ReportingTables.pdf

Energy

- V2_0_Cover.pdf
- V2_1_Ch1_Introduction.pdf
- V2_2_Ch2_Stationary_Combustion.pdf
- V2_3_Ch3_Mobile_Combustion.pdf
- V2_4_Ch4_Fugitive_Emissions.pdf
- V2_5_Ch5_CCS.pdf
- V2_6_Ch6_Reference_Approach.pdf
- V2_x_An1_Worksheets.pdf

IPPU

- V3_0_Cover.pdf
- V3_1_Ch1_Introduction.pdf
- V3_2_Ch2_Mineral_Industry.pdf
- V3_3_Ch3_Chemical_Industry.pdf
- V3_4_Ch4_Metal_Industry.pdf
- V3_5_Ch5_Non_Energy_Products.pdf
- V3_6_Ch6_Electronics_Industry.pdf
- V3_7_Ch7_ODS_Substitutes.pdf
- V3_8_Ch8_Other_Product.pdf
- V3_x_An1_Worksheets.pdf
- V3_x_An2_Potential_Emissions.pdf
- V3_x_An3_Improvements.pdf
- V3_x_An4_IPPU_Glossary.pdf

Waste

- V5_0_Cover.pdf
- V5_1_Ch1_Introduction.pdf
- V5_2_Ch2_Waste_Data.pdf
- V5_3_Ch3_SWDS.pdf
- V5_4_Ch4_Bio_Treat.pdf
- V5_5_Ch5_IOB.pdf
- V5_6_Ch6_Wastewater.pdf
- V5_x_An1_Worksheet.pdf

AFOLU

- V4_00_Cover.pdf
- V4_01_Ch1_Introduction.pdf
- V4_02_Ch2_Generic.pdf
- V4_03_Ch3_Representation.pdf
- V4_04_Ch4_Forest_Land.pdf
- V4_05_Ch5_Cropland.pdf
- V4_06_Ch6_Grassland.pdf
- V4_07_Ch7_Wetlands.pdf
- V4_08_Ch8_Settlements.pdf
- V4_09_Ch9_Other_Land.pdf
- V4_10_Ch10_Livestock.pdf
- V4_11_Ch11_N2O&CO2.pdf
- V4_12_Ch12_HWP.pdf
- V4_12_Ch12_HWP_Worksheet.xls
- V4_12_Ch12_HWP_Worksheet.zip
- V4_13_An1_Worksheets.pdf
- V4_14_An2_SumEqua.pdf
- V4_p_Ap1_Charcoal.pdf
- V4_p_Ap2_WetlandsCO2.pdf
- V4_p_Ap3_WetlandsCH4.pdf

Basic estimation methods

- $Emissions = Activity\ Data\ (AD) * Emission\ Factors\ (EF)$

The basic equation can in some circumstances be modified to include other estimation parameters than emission factors.

- IPCC Emission Factor Database:

  https://www.ipcc-nggip.iges.or.jp/EFDB/main.php

  It should be noted that the *IPCC 2006 Guidelines* also contain mass balance methods.

Estimation method for emissions from fossil fuel combustion

- If data on sectorial and fuel-specific activity data and EF are available

$$Emissions = \sum\sum(activity\ data_{i,j} \times EF_{i,j})$$

where, i is fossil fuel types, j is sectors.

- $EF_{i,j}$ can be modified to include other estimation parameters:

$$EF = v_{i,j} \times c_{i,j} \times o_{i,j}$$

where $v_{i,j}$ is net heating value, $c_{i,j}$ is carbon content, $o_{i,j}$ is oxidization rate.

If $v_{i,j}$ and $c_{i,j}$ are not directly available, the mass carbon content (C_{ar}) is usually used.

$$EF = C_{ar\,i,j} \times o_{i,j}$$

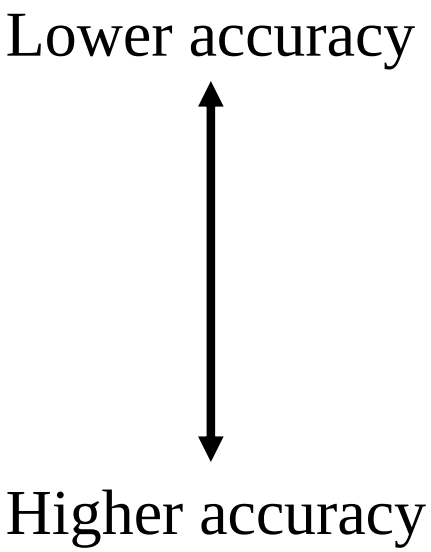
Tiered Methods

- Tiered methods help match method to data capacity

Tier 1: default methods and default emission factors

Tier 2: country-specific emission factors

Tier 3: detailed models, facility-level data, or high-resolution methods



In the 2006 IPCC Guidelines Vol.2 Energy, it presents a generalized decision tree for selecting Tiers for fuel combustion (Figure 1.2).

Table 1-1 Methodologies Used for NGI2020

Source/Sink Categories	CO_2		CH_4		N_2O	
	Methodology	Emission Factors	Methodology	Emission Factors	Methodology	Emission Factors
1.A.1 Energy industries	T1, T2	D, CS	T1, T2	D, CS	T1, T2	D, CS
1.A.2 Manufacturing industries and construction	T1, T2	D, CS	T1	D	T1	D
1.A.3 Transport	T1, T2, T3	D, CS	T1, T3	D, CS	T1, T3	D, CS
1.A.4 Other sectors	T1, T2	D, CS	T1	D	T1	D
1.B.1 Solid fuels			T1, T2, T3	D, CS		
1.B.2 Oil and natural gas system			T1, T3	D, CS		
2.A Mineral industry	T1, T2	D, CS				
2.B Chemical industry	T1, T2	D, CS			T1, T2	D, CS
2.C Metal industry	T1, T2	D, CS	T1	D	NO	NO
2.D Non-energy products from fuels and solvent use	T1	D				
3.A Enteric fermentation			T1, T2	D, CS		
3.B Manure management			T1, T2	D, CS	T1, T2	D, CS
3.C Rice cultivation			T2, T3	CS		
3.D Agricultural soils					T1, T2	D, CS
3.F Field burning of agricultural residues			T1	D	T1	D
4.A Forest land	T2	CS	T1	D	T1	D
4.B Cropland	T3	CS				
4.C Grassland	T2	CS	T1	D	T1	D
4.D Wetlands	T2	CS	T2	CS	NE	NE
4.E Settlements	T2	CS				
4.F Other land	T2	CS				
4.G Harvested wood products	T2	CS				
4.H Other biomass	T2	CS				
5.A Soil waste disposal			T2	D, CS		
5.B Biological treatment of solid waste			T1	D	T1	D
5.C Incineration of waste	T2	CS	T1, T2	D, CS	T1, T2	D, CS
5.D Wastewater treatment and discharge			T2	CS	T1	D
1.D.1.a International aviation	T3	CS	T3	CS	T3	CS
1.D.1.b International navigation	T1	D	T1	D	T1	D
1.D.3 CO_2 emissions from biomass	T1	D				

Table 1-1 continued

Source/Sink Categories	HFCs		PFCs		SF_6	
	Methodology	Emission Factors	Methodology	Emission Factors	Methodology	Emission Factors
2.A Mineral industry						
2.B Chemical industry	T1,T2	D,CS	T1	D	T1	D
2.C Metal industry			T2	CS	NO	NO
2.D Non-energy products from fuels and solvent use						
2.E Electronics industry			T2	CS		
2.F Product uses as substitutes for ozone depleting substances	T1,T2	D,CS				
2.G Other product manufacture and use					T2	CS

Notes: 1). The methodological codes T1, T2 and T3 represent Tier 1, Tier 2 and Tier 3 methods respectively.
2). The emission factor code CS represents the country-specific emission factor in China, D represents the default IPCC emission factor.
3). Shaded cells do not require entries.
4). "NE" (not estimated) indicates that a particular source or sink category has not been estimated.
5). "NO" (not occurring) indicates that a particular source or sink category do not occur.
6). When listed together, it means that subcategories within this category use different Tier methods or emission factor data sources.

$$Emissions = Activity\ Data\ (AD) * Emission\ Factors\ (EF)$$

Where does activity data come from?

Where do emission factors come from?

China. 4th BUR on Climate Change (2024)

From Administrative Statistics to GHG Inventory

- A national GHG inventory following IPCC method requires translating statistical data into emissions through:
  - **Reclassification**
    - Activities must be reorganized according to inventory sectors and source categories.
  - **Conversion**
    - Activity data must be converted into emissions using emission factors and carbon content.
  - **Boundary rules**
    - Only emissions/removals within the reporting boundary and accounting rules are included.
  - **Documentation and consistency**
    - Methods, assumptions, data sources, and uncertainties must be documented over time.

Discussion

Why the energy sector is central in national GHG inventories?

03

How Comparable and Reliable Are National Emissions Estimates?

Production vs Consumption-based Accounting;

Gigaton Gap Case discussion

Case Analysis:
Production-based vs Consumption-based

Case study: **Country A produces 1,000,000 tons of steel; 400,000 tons are consumed domestically, and 600,000 tons are exported to Country B. The production process emits a total of 5,000,000 tons of CO_2.**

国家A生产100万吨钢材，国内消费40万吨，出口60万吨至国家B。生产过程总计排放500万吨CO_2。

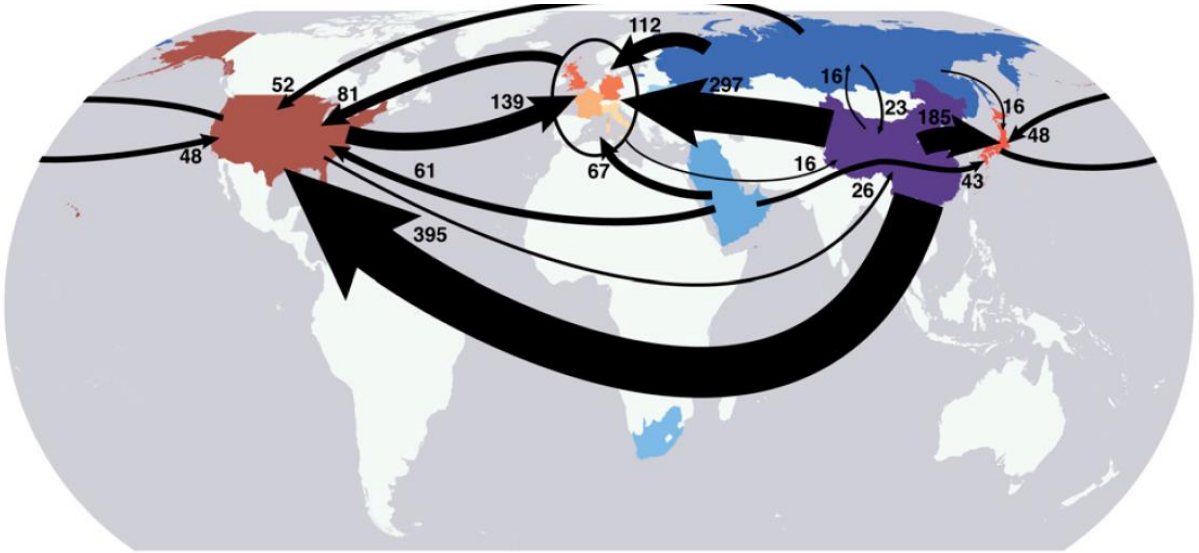
What are the GHG emissions for Country **A** and **B,** respectively, from both the production and consumption perspectives?

	Production-based accounting	Consumption-based accounting
Country A		
Country B		

What respective responsibility allocation principles do the two approaches embody?

National Level Carbon Accounting: PAP vs CAP

- National inventories are generally based on territorial /production-based logic:
  - Emissions are counted where they physically occur.
- National GHG emissions can also be counted towards what driven by consumption in this territory, that is following consumption-based logic.
  - Emissions associated with the goods and services consumed by the reporting entity (e.g. households within a country).



Source: Davis, S. J., & Caldeira, K. (2010). Consumption-based accounting of CO_2 emissions. *Proceedings of the National Academy of Sciences of the United States of America*, *107*(12), 5687–92. doi:10.1073/pnas.0906974107

Production-based accounting principle

- Production-based accounting principle (PAP)
  - Pros
    - Relatively certain/easy to quantify
    - Clear scope of “responsibility”
    - Good for legally binding targets
    - …
  - Cons
    - Focuses on a limited range of emission sources
    - Leakage – changes in emissions outside the inventory boundary are not shown
    - …

Consumption-based accounting principle

- Consumption-based accounting principle (CAP)
  - Pros
    - Avoids “off-shoring” or leakage
    - Reflects “responsibility” for emissions (if consumption is ultimate driver)
    - Covers a wider range of mitigation options?
    - Political or ethical advantages?
  - Cons
    - Higher uncertainty
    - Excludes exports
    - Others?

[Greta Thunberg says UK climate leadership is a 'lie‘]



Case Analysis:
The Gigaton Gap in China's CO_2 Inventories

$$AD_{\text{national}} \times EF \neq AD_{\text{provincial}} \times EF$$

Where Does the 1.4 Gt Gap Come From?

- China's CO_2 emissions in 2010

Based on national energy statistics

7.693 Gt CO_2

Based on the sum of provincial statistics

9.084 Gt CO_2

A gap of 1.391 Gt CO_2

Both estimates use official energy statistics and follow the IPCC accounting approach.

Why are they so different?

投票 最多可选1项

设置

Where would you look first?

A Accounting subject

B Boundary and attribution

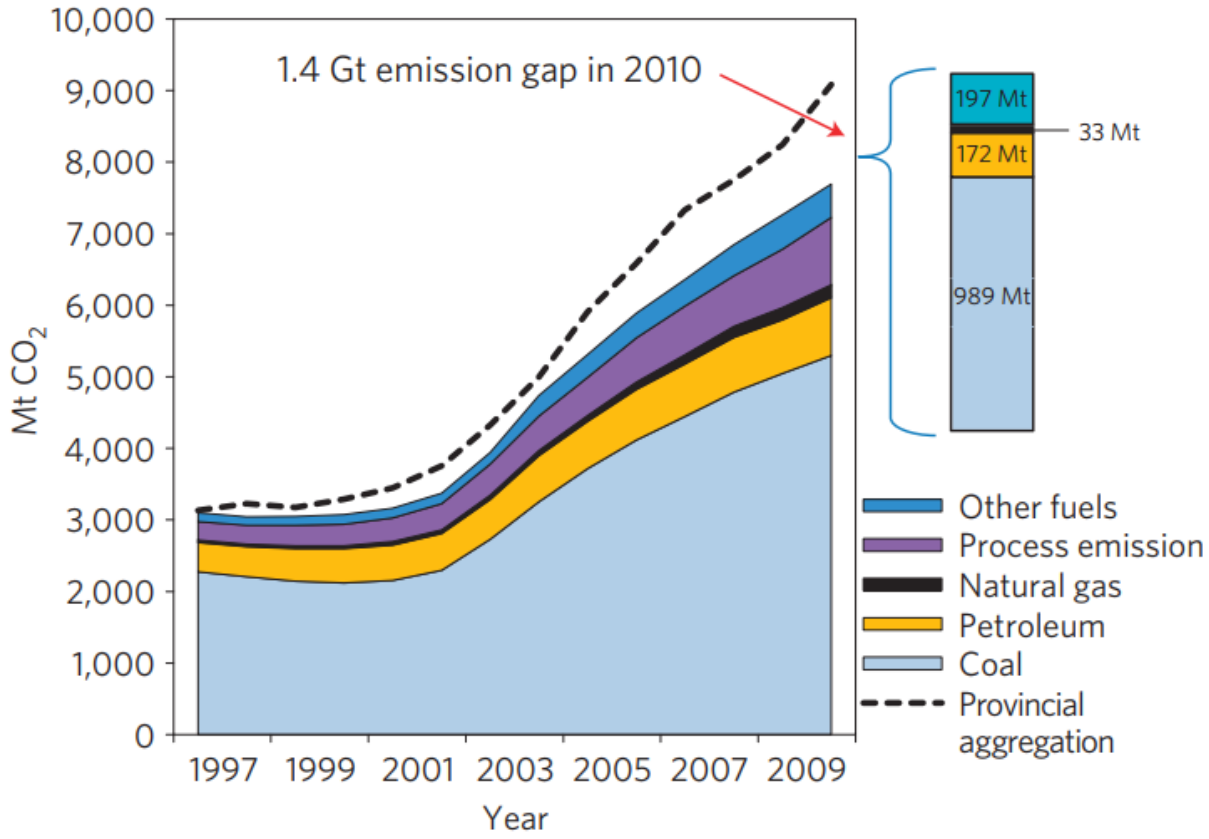
C Activity data

D Emission factors

E Calculation error

提交

Clue 1: Look at the pattern



The area chart (left) illustrates the increase of CO_2 emissions calculated from the national energy statistics since 1997 broken down by different fuel type. Other fuels include, for example, coke oven gas, other gas, other coking products, LPG, refinery gas and other petroleum products. The dashed line represents the aggregated CO_2 emissions calculated from the provincial energy statistics 1997–2010. **The column chart (right) presents the 1.4 Gt emission gap in 2010 between national and provincial statistics** and the pattern of different fuel types in contributing to the emission gap.

Figure 1 | The sources of China's CO_2 emissions by fuel type during 1997–2010.

From: Guan et al. (2012)

单选题 1分

设置

What does the time series suggest?

A The discrepancy appeared suddenly in 2010.

B It was a one-off calculation error.

C The two statistical series had diverged for many years.

D The accounting boundary changed every year.

提交

Clue 2: Follow the coal

- Evidence from 2010 (Guan et al., 2012):
  - 71% of the emissions gap was associated with coal.
  - National raw-coal consumption: 3,163 Mt
  - Provincial sum of raw-coal consumption: 3,910 Mt

⟶ A 24% difference in the underlying coal activity data

Different activity data
→ different emissions estimates
→ concerns about **consistency** and **uncertainty**

多选题 1分

设置

Which conclusions are directly supported by the study?
Select all that apply.

A The same estimation method can produce different results when the activity data differ.

B Much of the discrepancy is associated with coal-consumption statistics.

C The provincial estimate is the true value of China’s emissions.

D National and provincial energy statistics require further reconciliation and validation.

E The national estimate is false.

提交

F The study proves whether China met its carbon-intensity target.

Supported by the paper

- Two official energy datasets produce different emissions estimates
- **Coal data** explain much of the gap
- Data consistency and validation need improvement
- The discrepancy existed before 2010

Not established by the paper

- Which estimate is the “true” national total
- That one dataset is simply correct and the other wrong
- Whether China achieved its official intensity target
- That the gap suddenly appeared in 2010

Why did the two energy statistics diverge?

Why Can National Emissions Results Differ?

Case 1: PBA vs CBA

What is the key question?

- Which emissions should be attributed to a country?

Why do the results differ?

- Different accounting boundaries
- Different attribution rules
- Emissions embodied in trade are treated differently

What should we examine?

- Purpose and intended use
- Boundary and attribution
- Meaning and comparability of the results

Case 2: Gigaton Gap

What is the key question?

- Why can estimates of the same country's territorial emissions disagree?

Why do the results differ?

- Different underlying energy statistics
- Inconsistencies between national and provincial data
- Uncertainty in activity data

What should we examine?

- Data sources and quality
- Completeness and consistency
- Uncertainty and verification

Different results do not always mean that one result is wrong.
They may reflect different accounting questions—or different evidence.

04

Wrap-up and Next Time

Wrap up – understanding a national GHG inventory

饮 水 思 源 · 爱 国 荣 校

Why – why is it produced?

- Reporting
- Targets
- Policies
- …

What – what is counted?

- Territory or Consumption
- Gases, sectors
- Reporting year

How – how is it quantified?

- IPCC guidelines
- Activity Data * Emission Factors
- Statistics/Data

So What – how to interpret it?

- Comparability
- Model/Method choices
- Data differences

Before comparing two emissions figures, ask: Do they count the same thing—and are they based on comparable data?

NEXT: What changes when the accounting subject becomes a city?

Next Time – Organizational Level Carbon Accounting

- **Essential reading:**
  - **WRI, C40 and ICLEI. *Global Protocol for Community-Scale Greenhouse Gas Inventories, Version 1.1*.** Read the Executive Summary (page 9-page 15), and the parts of Chapter 4 explaining Scopes 1–3 and BASIC/BASIC+ (page 39-page 41).
- **Recommended reading:**
  - **World Bank (2021). *GHG Emissions Inventories: An Urban Perspective*.** Read pp. 11–14.
- **Further reading:**
  - **Chen, G. et al. (2019). "Review on City-Level Carbon Accounting."** Read the Abstract, Introduction and Conclusion.

While reading, focus on one question:
How does the GPC account for emissions when a city's activities extend beyond its geographic boundary?





See you next time

Have a good holiday!